

Vision-Guided Synchronization of Robot Motion and Material Flow in Pellet Extrusion

PhD Research Proposal (Draft)

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Abstract

Long-duration robotic extrusion offers a pathway to high-throughput architectural fabrication but is currently limited by geometric drift arising from the coupled dynamics of robot motion, extrusion mechanics, and time-variant material states. This research proposes a closed-loop framework for pellet-screw extrusion that synchronizes robot velocity with material flow through in-situ vision. The dissertation is structured around three contributions. Paper 1 establishes the closed-loop system: a Proportional and Integral (PI) controller adjusts Tool Center Point (TCP) velocity via the KUKA Robot Sensor Interface (RSI) interface based on vision-derived bead width estimates, and its performance is validated against open-loop baselines on progressively complex geometries over multi-hour builds. Paper 2 benchmarks surface texture against bead geometry as an indicator of extrusion instability, measuring whether texture-based features detect flow perturbations earlier than geometric ones, and evaluates early and mid-level fusion of both modalities. Paper 3 tests cross-polymer transferability of the trained perception pipeline by applying the model zero-shot to materials with distinct rheological behavior and quantifying the labeled data required to recover performance. Together, these contributions aim to replace human-in-the-loop oversight with a vision-guided control architecture that extends the feasible duration, complexity, and material scope of architectural-scale Additive Manufacturing (AM).

Background

Contemporary architecture increasingly demands non-standard, geometrically complex components, such as curved façades, topology-optimized structural nodes, and bespoke formwork, which conventional mold-based construction cannot economically produce at scale (Menges et al., 2017). This geometric ambition, coupled with the sector’s persistent productivity gap and high material waste (Barbosa et al., 2017), drives the transition toward digital fabrication (DF), translating computational design directly into physical form through automated, data-driven production (Gramazio & Kohler, 2014). Within the DF framework, Robotic Additive Manufacturing (RAM) provides the multi-axis agility required for architectural-scale components (Keating et al., 2017). Unlike static 3-axis systems, RAM leverages robotic flexibility to produce complex, non-planar geometries without formwork (Helm et al., 2014). Pellet-screw extrusion extends these capabilities by delivering the high throughput required for large structures (Roschli et al., 2019). In practice, large-format polymer extrusion in Architecture, Engineering, and Construction (AEC) field serves two primary roles: producing complex formwork for casting concrete or other composites,

and fabricating end-use objects such as furniture, sculptural installations, and landscape elements (Jipa & Dillenburger, 2022; Vicente et al., 2023). Demonstration projects in large-format polymer extrusion, such as printed pavilion molds exceeding two meters and freestanding furniture with double-curved surfaces, confirm the architectural potential of this pipeline, but also reveal that geometric fidelity degrades over multi-hour builds when process parameters remain static (Post et al., 2019). However, the transition to long-duration builds introduces process instabilities. Environmental changes and material fluctuations cause geometric drift that static parameters cannot correct (Bellini et al., 2004; Pricci et al., 2021). Maintaining deposition fidelity over extended periods therefore requires real-time sensing and closed-loop control, a challenge that motivates the current research. Fig. 1 illustrates this challenge: a bio-based façade panel produced by the author required continuous manual adjustment throughout ten-hour robotic extrusion.



Figure 1: Mycelium-mineral composite façade panel produced through robotic extrusion with space-filling curve toolpaths in prior work by the author’s group. The build required continuous human oversight to maintain deposition quality, illustrating the reliance on manual intervention that currently limits the scalability of architectural-scale digital fabrication.

Pellet extrusion sits at the intersection of three coupled subsystems: the robot manipulator, the single-screw extruder, and the deposited polymer (Vicente et al., 2023). Standard workflows treat these components as decoupled and assume the system functions as a deterministic actuator (Turner et al., 2014). In practice, this setup often relies on a human-in-the-loop strategy where operators manually adjust parameters through ad-hoc overrides (Chesser et al., 2019; Vera-Sorroche et al., 2014). However, manual intervention cannot match the millisecond-level dynamics of robotic motion (Go et al., 2017). Unlike filament-fed systems, the pellet extruder’s output is mediated by screw geometry, barrel thermal zones, and melt pressure (Pricci et al., 2021). This paradigm suffices for short desktop prints where thermal transients are brief and bead dimensions are small (Brenken et al., 2018). At the scale of large-format architectural fabrication, beads are an order of magnitude wider and build volumes span several cubic meters (C. E. Duty et al., 2017). Under these conditions, small perturbations accumulate into significant structural and geometric defects over multi-hour build horizons (Post et al., 2019).

The polymer feedstock undergoes a complete thermomechanical cycle: solid pellets are conveyed, compressed, and melted inside the barrel; the resulting melt is pumped through the nozzle; and the deposited bead cools and solidifies (Calafel et al., 2020). This process is governed by temperature- and shear-dependent material parameters (Rauwendaal, 2001). Polymer melts are predominantly shear-thinning, which means screw speed changes do not produce proportional changes in volumetric output (Tadmor & Gogos, 2006). The flow response is also material-specific. Semi-crystalline polymers like PLA exhibit a sharp solid-to-melt transition at their melting temperature (T_m), whereas amorphous polymers like ABS soften gradually above their glass transition temperature (T_g) (C. Duty et al., 2018). These differences lead to distinct melting efficiencies and flow behaviors under the same thermal input (La Gala et al., 2021). On the deposition side, the cooling rate governs both dimensional stability and interlayer bond quality. In semi-crystalline polymers, slower cooling promotes higher crystallinity and greater volumetric shrinkage, while the bead must remain above a critical temperature long enough for molecular

diffusion to form a cohesive weld with the previous layer (Seppala & Migler, 2016). At architectural scales, larger bead cross-sections and longer inter-layer intervals amplify these thermal gradients, making cooling-induced distortion a persistent source of cumulative geometric deviation (Compton et al., 2017). Furthermore, prolonged heating and batch-to-batch variation in pellets introduce stochastic disturbances to melt consistency (Abeykoon, 2016). These material drifts alter the system transfer function over time, requiring real-time perception for stable process control (Abeykoon, 2014).

The synchronization of machine motion and material flow depends on the alignment between the robot and the extruder (Go et al., 2017). A fundamental mismatch exists because the robot operates with millisecond-level precision, while the pellet extruder responds with latency caused by melt compressibility and pressure buildup (Chesser et al., 2019). The extrusion dynamics can be approximated by a First-Order-Plus-Dead-Time (FOPDT) transfer function,

$$G(s) = \frac{Y(s)}{U(s)} = \frac{Ke^{-Ls}}{\tau s + 1},$$

where $U(s)$ and $Y(s)$ denote the command input and the actual volumetric output in the Laplace domain respectively. K is the process gain, τ is the time constant governing the response speed, and L is the dead time representing the delay between actuation and material deposition (Abeykoon, 2016). This model captures both the gradual flow response of the extruder and the latency that causes temporal mismatch with robot motion. However, the G-code schedule typically treats the relationship between input and output as static and time-invariant (Turner et al., 2014). During long-duration builds, the parameters K and L drift as thermal fluctuations and material degradation alter melt viscosity and flow behavior (Vera-Sorroche et al., 2014). This evolving state desynchronizes the fixed G-code command from the actual material response, producing over-extrusion or under-extrusion (Greeff & Schilling, 2017). Over multi-hour horizons, such parameter drift accumulates into geometric deviation and structural defects that compromise the integrity of architectural-scale components (Ertay et al., 2018).

Closing this loop requires a perception system estimating process state at a suitable rate for control. In extrusion-based additive manufacturing (AM), current monitoring mainly relies on computer vision of geometric features such as bead width, layer height, and cross-sectional profile, which are useful for post-hoc quality assessment and batch-level parameter adjustment (Oleff et al., 2021). However, geometric features are lagging indicators because they record the accumulated effect of upstream flow disturbances rather than their onset (Oleff et al., 2021). This limitation is especially relevant at architectural scale, where printed formwork and freestanding objects need to maintain macro-geometric fidelity throughout multi-hour builds to avoid delamination, warping, and toolpath drift (Jipa & Dillenburger, 2022; Vicente et al., 2023). Surface-level finish is often less critical during deposition because it can be improved later through post-processing (Post et al., 2019). For this reason, in-process monitoring should detect flow perturbations before they accumulate into macro-scale geometric failure. In polymer extrusion, surface texture provides an earlier signal because extrusion instabilities appear as visible surface patterns linked to the rheological stress state at the die exit (Agassant et al., 2006). Instabilities, such as sharkskin, melt fracture, and stick-slip are examples of this behavior (Agassant et al., 2006). Fig. 2 illustrates the progression of these surface defects with increasing shear rate. Because these texture patterns can emerge before measurable changes in bead width or layer height, they can reveal flow-rate perturbations earlier than geometry-based metrics (Vergnes, 2015). Related AM fields have already used real-time visual feedback for process compensation, including vision-based extrusion regulation in robotic concrete printing (Kazemian et al., 2019). Multi-sensor fusion has also been used for defect prediction in laser-directed energy deposition (Chen et al., 2023). However, no comparable framework has been established for pellet-screw polymer extrusion, where viscoelastic behavior, long build durations, and the need for multi-material generalization create different constraints (Pricci et al., 2021). This dissertation aims to address that gap by benchmarking texture against geometry, fusing both modalities, and integrating a local process model for real-time adaptation.

This dissertation draws on three intersecting domains: large-format robotic additive manufacturing, which defines the physical platform and architectural-scale constraints; in-situ visual sensing, which

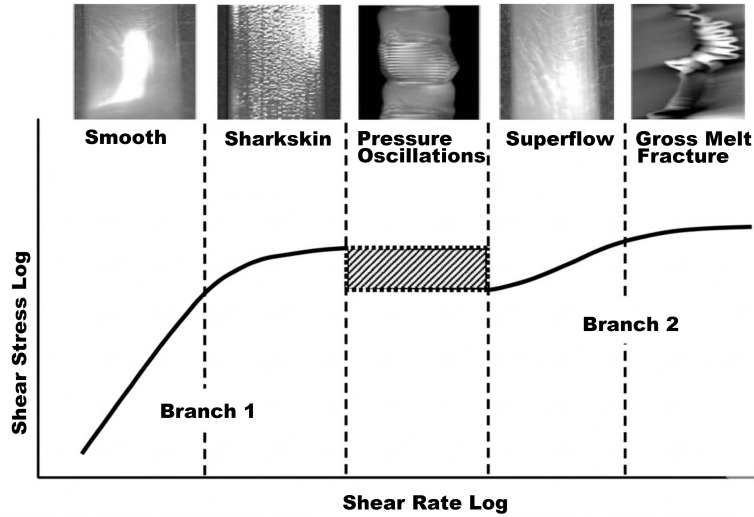


Figure 2: Extrusion instability progression in polymer melts as a function of increasing shear rate, with the corresponding flow curve showing wall shear stress versus shear rate. At low shear rates (Branch 1), the extrudate surface is smooth (I). As shear rate increases, surface texture degrades through sharkskin (II) and pressure oscillations (III) before a brief return to smooth extrusion in the superflow regime (IV). At high shear rates (Branch 2), gross melt fracture (V) distorts the entire extrudate volume. The surface signatures can precede measurable geometric deviation. Adapted from Agassant et al. (2006).

provides the real-time state estimates required for closed-loop control; and data-driven process modeling, which maps perceptual signals to generalizable defect predictions across materials. This dissertation targets at integrating all three into a closed-loop framework that synchronizes robot motion with material flow under visual guidance. Establishing this closed-loop capability carries direct impact on the constructed environment. In formwork applications, the dimensional tolerance of the printed mold propagates into the cast element. Uncorrected drift during long-duration prints can cause dimensional deviations that exceed the assembly tolerances required for façade panels and structural connections (Jipa & Dillenburger, 2022). For freestanding printed components, cumulative over-extrusion alters cross-sectional mass distribution, compromising both structural performance and design intent (C. E. Duty et al., 2017). More broadly, if process stability can only be guaranteed for short, geometrically simple toolpaths, designers are forced to partition complex forms into small segments and manually oversee assembly, negating the scalability promise of robotic fabrication (Keating et al., 2017). A reliable closed-loop framework therefore does not merely solve a manufacturing defect, but it expands the feasible design space for architectural-scale additive manufacturing by enabling longer, more complex, and less supervised build process.

Problem Statement

As aforementioned, pellet-screw extrusion is governed by coupled and time-variant dynamics: the extruder's FOPDT response drifts as thermal conditions and melt viscosity evolve, while the robot executes commands with negligible latency. Over multi-hour builds, this mismatch accumulates into geometric deviation where G-code or manual overrides cannot correct in time (Chesser et al., 2019; Greeff & Schilling, 2017). Existing automated approaches address fragments of this problem but fall short in three dimensions: temporal scope, perceptual coverage, and algorithmic generalization.

On the temporal dimension, vision-based monitoring in extrusion AM operates at three levels of granularity. The majority of published work is *post-print*: models are trained on images collected after a build is complete, then used to adjust parameters for subsequent prints (Delli & Chang, 2018). A growing body of work performs *layer-wise* inspection, where the nozzle pauses between layers for image capture and defect classification (Baumann & Roller, 2016; W. Wang et al., 2024). YOLO-based detectors have

been applied to classify over-extrusion, under-extrusion, and stringing from top-down layer photographs (W. Wang et al., 2024), but pausing between layers prevents continuous tracking of within-layer drift. The scarce setup is *continuous in-process* monitoring. The only example employs an eye-on-nozzle camera and a multi-head neural network, achieving cross-geometry and cross-material error detection and correction in real time on desktop Fused Filament Fabrication (FFF) printers (Brion & Pattinson, 2022). However, the experimental scope was limited to small desktop prints lasting minutes, on filament-fed systems with fast thermal equilibration. Whether such regulation remains stable over multi-hour, large-format builds has not been tested. In pellet-screw extrusion, thermal inertia and melt compressibility introduce longer dead times and greater hysteresis (Abeykoon, 2016). Such continuous regulation has not been validated.

On the perceptual dimension, current monitoring relies mostly on geometric features: bead width, layer height, or cross-sectional profile (Oleff et al., 2021). Geometry is a lagging indicator recording the accumulated result of flow disturbances rather than their onset (Vergnes, 2015). Surface texture, such as sharkskin and melt fracture, reflects the rheological stress state at the die exit and can precede measurable geometric change (Vergnes, 2015). Yet texture has not been used as an explicit feedback signal in extrusion-based AM. Where texture information is captured, it enters indirectly through end-to-end models learning mixed representations without distinguishing geometric from textural cues (Brion & Pattinson, 2022). No published work has benchmarked texture against geometry as a predictive feature for process state estimation. Separately, multi-sensor fusion has matured in other AM domains: in Laser Powder Bed Fusion (LPBF), the combination of thermal imaging, acoustic emission, and melt-pool vision has been reviewed extensively (Papatheodorou et al., 2025). In material extrusion, fusion work is scarce. A recent study fused multiple sensor modalities for anomaly detection in FFF using contrastive causal alignment (Gu et al., 2025). But the fusion was operated at the decision level rather than integrating complementary physical signals into a process state estimate. No comparable framework exists for pellet-screw polymer extrusion.

On the algorithmic dimension, the prevailing approach treats extrusion defect detection as a generic object detection or image classification task, typically fine-tuning YOLO variants or lightweight CNNs on self-collected datasets with categorical labels assigned by hand (W. Wang et al., 2024). A recent research pursued a different route, using *process-parameter deviation* as the label: each image was tagged with commanded flow rate, speed, Z-offset, and temperature, and the model learned to predict these continuous values from visual appearance (Brion & Pattinson, 2022). This automated labeling avoids manual annotation and scales to large datasets. However, it rests on an assumption breaking down in pellet extrusion: a given commanded input produces a deterministic visual outcome. In filament-fed desktop printing, this assumption holds because the feedstock is mechanically metered and thermal response is fast (Turner et al., 2014). In pellet-screw extrusion, the same commanded screw speed can produce different visual signatures depending on hidden state variables. For instance, elevated barrel temperature may yield uniform over-extrusion, whereas wall slip at the die exit may produce sharkskin texture, and excessive die-exit stress may trigger melt fracture (Agassant et al., 2006; Vergnes, 2015). Which outcome occurs depends on instantaneous melt temperature, shear history, and molecular-weight distribution (C. Duty et al., 2018; Tadmor & Gogos, 2006). This one-to-many mapping between commanded parameters and actual process states means that parameter-based labels carry systematic ambiguity, limiting both classification accuracy and the model’s ability to prescribe correct interventions. More broadly, few-shot learning and domain adaptation have gained traction in general manufacturing inspection (Zajec et al., 2024), but remain largely unvalidated in extrusion-based AM. Self-supervised pretraining and Vision Transformers have begun to appear in isolated studies on desktop-scale FFF (X. Q. Wang et al., 2025), yet the question of how to train a perception model that transfers across polymer types with rheological signatures has not been addressed.

The result is a gap at the intersection of these three dimensions. No existing framework provides continuous, in-process synchronization of machine motion with material deposition over multi-hour builds, while fusing texture and geometry for state estimation, and generalizing across materials through a transferable process model. This dissertation targets that gap: building a perception-driven control architecture that operates in real time at architectural scale, benchmarks complementary visual modalities,

and validates cross-material transfer. If process stability no longer depends on short toolpaths and constant human oversight, designers can commit to geometries that require uninterrupted multi-hour deposition, such as large continuous-shell formwork and monolithic freestanding elements, without sacrificing dimensional control.

Research Questions

- **RQ1:** What are the limits of geometric fidelity in long-duration robotic pellet extrusion, and how does closed-loop control extend them?
- **RQ2:** Under what conditions does surface texture precede geometric deviation as an indicator of extrusion instability?
- **RQ3:** What determines whether a vision-based process model trained on one polymer remains valid for others?

Objectives

1. **Closed-Loop System Validation** Build a vision-based closed-loop system for real-time robot speed adjustment. The validation process uses PLA to print cylinders, tapered forms, and twisted columns. Each build lasts at least two hours. A direct comparison between open-loop and closed-loop outcomes identifies the specific stages and geometries where active control improves dimensional accuracy.
2. **Detection Latency: Texture vs. Geometry** Compare surface texture and bead geometry as indicators of extrusion failure. Controlled experiments will trigger sharkskin, melt fracture, and inconsistent extrusion. Measuring the time gap between simultaneous signals across various polymers determines when texture offers a faster, more effective early warning.
3. **Model Generalization and Failure Analysis** Evaluate model performance on new materials without additional training. The test uses polymers with distinct flow behaviors, such as ABS and carbon-fiber PLA. This study quantifies accuracy loss and the data volume required for performance recovery. The final analysis identifies whether failures stem from visual features, material physics, or the model architecture itself.

Setup

Hardware

The primary platform is a KUKA KR10 R1100-2 (6-axis, reach 1100 mm, repeatability ± 0.03 mm). The pellet-screw extruder is Massive Dimension MDPH2, fitting with a 3 mm nozzle. The print bed is a custom frame assembled from standard 1×2 in steel tubing, surfaced with 3/4 in Medium-density Fiberboard (MDF), and clamped to a steel welding table. An eye-in-hand ZED2i stereo camera (Stereolabs) is mounted on the robot's end effector approximately 400 mm from the nozzle tip. This distance places the nozzle at the edge of the camera's reliable depth range (manufacturer spec: 0.2 m; practical minimum for accurate depth: approximately 0.5 m). Depth accuracy at this working distance will be characterized in Paper 1, and if insufficient for bead-height measurement, a supplementary structured-light sensor at closer range will be evaluated.

The current mounting position provides a clear view of the deposited bead behind the nozzle (trailing view), but the nozzle body partially occludes the bead ahead of the travel direction. Paper 1 will determine whether trailing-view data alone is sufficient for closed-loop control. The complete hardware configuration is shown in Fig. 3. If forward-looking coverage proves necessary, two options will be evaluated: adding

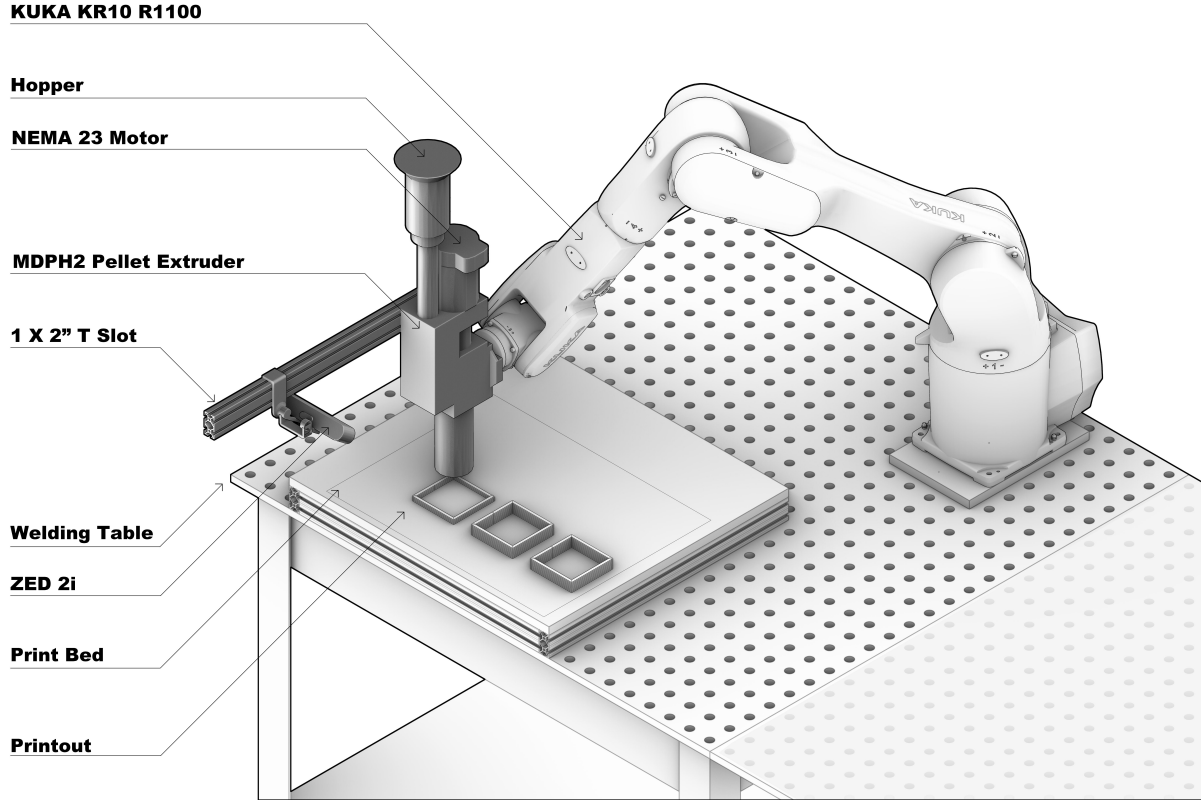


Figure 3: Hardware setup. KUKA KR10 R1100-2 with Massive Dimension MDPH2 pellet-screw extruder and eye-in-hand ZED2i stereo camera mounted approximately 400 mm from the nozzle tip via a T-slot extension arm. The MDF print bed is clamped to a steel welding table. Printed specimens on the bed illustrate a typical multi-part layout during calibration runs.

a secondary close-range camera (e.g., a borescope-type module, as recently demonstrated for real-time nozzle monitoring in desktop FFF (Kondrat et al., 2025)) or adding a fixed eye-to-hand camera at a complementary angle.

All algorithm development and model training are conducted on the KR10 platform. A scale-up validation on a larger-reach KUKA manipulator, KR120 R2700 in the Allied Facility, is planned as the final experimental step. The specific model will depend on equipment availability at the time of testing.

Signals

Three signal streams are recorded during every print:

Robot state. Tool Center Point (TCP) position, velocity, and orientation are streamed at the KUKA Robot Sensor Interface (RSI) cycle rate. TCP velocity serves as the primary control variable for closed-loop regulation. The extruder screw speed is set at a constant at the beginning during runs and corrected by the local classifier via RSI.

Vision. The ZED2i captures frames at 2K resolution (2208×1242 px for single side). A 224×224 px region of interest is cropped adjacent to the nozzle tip for bead analysis. At the current working distance the printed bead width occupies approximately 15-20 px in the Region of Interest (ROI). This pixel budget is sufficient for binary state classification (normal vs. anomalous) and coarse width estimation, but may limit the resolution of continuous width measurement. Paper 1 will quantify the effective mm/pixel ratio

and determine whether it constitutes a perception bottleneck for the controller, or whether the ROI size or camera position needs adjustment.

Thermal and environmental. Barrel zone temperatures are logged via Infrared camera for each print. Ambient lab temperature is recorded per session. Polymer grade, manufacturer, and melt flow index are documented for each material batch.

Factors Affecting Dimensional Accuracy

Several system-level parameters influence both the printed output and the perception measurements. All will be characterized and reported in Paper 1 to establish the error budget of the platform before proceeding to the perception and transfer studies in Papers 2 and 3.

- *Bed leveling.* The MDF surface is not precision-machined. Flatness variation across the print area will be measured (e.g., by probing with the robot tool tip) and reported relative to the 1.5 mm nominal layer height. Bead width depends on nozzle-to-substrate standoff, so bed-level error in the first layer propagates upward.
- *Robot path accuracy under load.* The manufacturer repeatability spec (± 0.03 mm) describes unloaded point-to-point return. Path accuracy during continuous motion with the extruder mounted is larger and will be measured separately. Based on the use experience, the braking distance (± 1 mm) will be used in first.
- *Vision pipeline latency.* The total time from frame capture through segmentation to control output determines the controller's effective sampling rate. Combined with robot travel speed, this latency defines the spatial distance the nozzle travels between sensing and correction.
- *Pixel resolution.* The mm/pixel ratio at working distance determines the smallest bead-width change the vision system can resolve. This will be calibrated and reported alongside the controller's performance to separate perception limits from control limits.
- *Extrusion thermal drift.* Barrel temperature fluctuation during multi-hour builds is the primary source of the FOPDT parameter variation as aforementioned. Logging thermal profiles allows correlation between temperature drift events and observed geometric deviation.
- *Material batch variation.* Polymer grade and batch are documented for each material used. Their Melt Flow Index (MFI) will be reported accordingly. The initial material is LX175 PLA.

Research Plan

The dissertation is structured as three papers. Each paper produces experimental artifacts and datasets that the subsequent paper builds on. Shared resources across all three papers include: baseline open-loop G-code schedules for each test geometry, full process logs during every print, and 3D deviation scans of all printed artifacts against their design models.

Test Geometries

The test geometries follow a sequence from isolated 2D toolpath features to multi-layer builds to volumetric forms. This change allows each level of geometric complexity to be validated before introducing additional variables such as thermal accumulation.

Single-layer toolpath features: Three planar features isolate distinct control challenges: straight segments, sharp corners with different degrees, and curved arcs. These single-layer prints serve as the baseline in Paper 1 to tune the controller before multi-layer builds.

Multi-layer short builds: The same three toolpath features are stacked for a limited number of layers to introduce inter-layer thermal effects and Z-direction error accumulation while keeping build duration short enough to isolate these effects from long-duration thermal drift.

Volumetric geometries: Three closed-contour forms are used for long-duration builds exceeding two hours: a straight-wall cylinder (approximately 150 - 200 mm diameter, 250 - 400 mm height), a tapered form, and a twisted column. The cylinder holds toolpath conditions constant across layers and isolates long-duration drift. The tapered form introduces a continuously changing target width. The twisted column combines sustained curvature change with direction reversals. At a layer height of 1.5 mm and typical deposition speeds, each geometry requires approximately 150 to 250 layers and 2 to 5 hours of continuous printing. Each volumetric geometry is printed in both open-loop and closed-loop modes, producing paired physical artifacts for demonstrators. The same sequence is repeated across materials in Paper 3 to hold geometry constant while varying rheology. The final experimental step is a scale-up print on the KR120 R2700 (see Setup) to test whether the framework transfers to different robot dynamics at a larger build volume.

Paper	Aim	Key Metrics
Paper 1	Geometric fidelity over long builds	Drift slope
Closed-loop fidelity	Effect of vision-guided velocity regulation	Settling time
		Defect rate
		As-built deviation
Paper 2	Conditions under which texture precedes geometry as a state indicator	Detection time gap
Texture vs. geometry	Labeling strategy comparison	F1 per process state
		Pipeline latency
Paper 3	Transferability of perception model across polymers	Zero-shot F1 drop
Cross-polymer transfer	Diagnosis of transfer failure modes	Samples to recover performance

Paper 1: Closed-Loop Fidelity

This paper characterizes how geometric fidelity degrades over long builds and across toolpath complexities, and quantifies how much vision-guided velocity regulation recovers.

System identification and control: Step-response tests on screw speed at multiple barrel temperatures quantify the FOPDT parameters (K , τ , L) and their thermal dependence. A PI controller with delay compensation adjusts TCP velocity via RSI to keep bead width within a target range. Screw speed is held constant. When the vision system detects deviation, the robot velocity is adjusted to match the observed material flow. TCP velocity is chosen as the control variable because the robot responds within milliseconds, whereas screw speed changes propagate through the FOPDT dynamics with a dead time on the order of seconds. Adjusting velocity rather than screw speed therefore provides the actuation bandwidth needed for real-time correction: slowing the robot when the observed bead is narrower than the target allows more material to accumulate per unit length, and accelerating it when the bead is wider reduces material per unit length. The signal architecture for both data collection and closed-loop deployment is summarized in Fig. 4. Two width reference strategies will be evaluated: (1) an absolute reference, comparing the estimated bead width against the target slice width from the toolpath; and (2) a relative reference, comparing the current frame’s width estimate against a rolling average of recent frames and triggering correction only when the deviation exceeds a threshold. The first requires accurate pixel-to-millimeter calibration; the second avoids this dependency by detecting change rather than measuring absolute dimension. Paper 1 will report which strategy is viable given the system’s pixel resolution and which produces more stable control.

Experiments: Following the progressive geometry sequence: (1) single-layer straight segments for step-response tuning, corners and arcs for settling time measurement; (2) multi-layer stacking of the same features to test controller robustness under inter-layer thermal effects without re-tuning; (3) volumetric long-duration prints (cylinder, tapered form, twisted column), each printed once open-loop and once

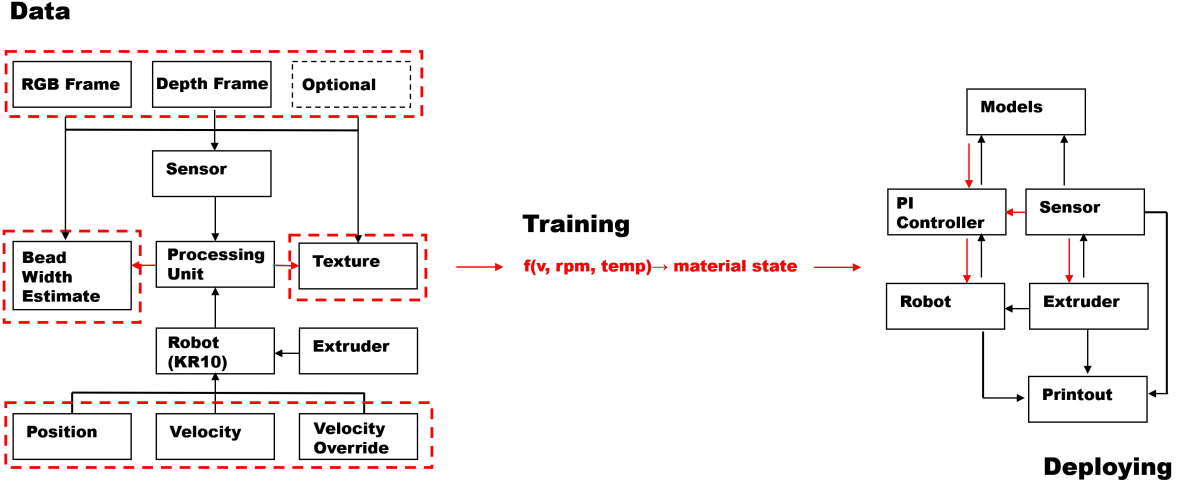


Figure 4: Signal architecture for data collection (left) and closed-loop deployment (right). During data collection, the ZED2i stereo camera captures RGB and depth frames while robot state (TCP position, velocity, and velocity override) and extruder parameters are logged synchronously. The processing unit extracts bead width estimates and texture features, which are used to train a mapping from process parameters to material state. During deployment, the trained model feeds a PI controller that adjusts TCP velocity via RSI to maintain target bead geometry. Screw speed remains constant throughout.

closed-loop, with all process data recorded for both evaluation here and retrospective analysis in Paper 2.

Contingency: If the PI controller with delay compensation proves insufficient for stable regulation, for example, if nonlinear flow dynamics cause persistent limit cycling, A model predictive control (MPC) formulation using the identified FOPDT model will be evaluated as an alternative. If depth accuracy at the current ZED2i working distance is inadequate for bead-height measurement, a supplementary structured-light sensor at closer range will be integrated, as noted in the hardware description. In either case, the downstream perception studies in Papers 2 and 3 remain valid because they depend on the vision classification pipeline, not on the specific control law.

Metrics:

- *Settling time:* standard control metric; time for bead width to return to tolerance after a transient.
- *Defect rate:* standard AM quality metric; percentage of toolpath length outside the tolerance band.
- *As-built deviation:* standard large-format AM practice; RMS surface error from 3D scan vs. design model, segmented by geometry region.
- *Drift slope:* defined here as the linear regression rate of bead width deviation over time (mm/hr); introduced because existing literature describes long-duration drift qualitatively without a standardized rate measure.

Paper 2: Texture vs. Geometry

This paper determines under what conditions surface texture detects extrusion instabilities before geometric deviation becomes measurable, and evaluates how labeling strategy affects classification accuracy.

Induced-failure experiments: Controlled perturbations push the extrusion system into known failure modes: gradually lowering barrel temperature to trigger sharkskin and then melt fracture, raising screw speed to induce over-extrusion, and reducing screw speed to produce starving. During each sequence, both texture features (from the near-nozzle ROI) and geometric features (bead width, height) are recorded simultaneously. Labels are assigned in two ways. First, because the perturbation schedule is known, each recording segment carries a coarse process-state label derived from the experimental condition. To reduce

manual annotation, self-supervised representation learning is used to extract frame-level embeddings from unlabeled print footage, and a lightweight classifier is fine-tuned on a small labeled subset. This approach has been validated for extrusion quality classification in desktop FFF (X. Q. Wang et al., 2025), though its effectiveness on pellet extrusion textures remains to be tested. A random subset of the auto-assigned labels is verified by visual inspection. Second, as a baseline comparison, each frame is also tagged with the commanded screw speed and temperature at the time of capture, reproducing the parameter-based labeling strategy where the model learns to predict process parameters from visual appearance. Comparing classifier performance under both labeling strategies isolates whether accuracy differences between modalities are genuine or an artifact of label ambiguity.

Modality benchmarking: Three feature sets are compared on the same dataset: geometry-only, texture-only, and fused. Geometry features consist of bead width and height estimated from the depth and RGB frames. Texture features capture surface regularity within the near-nozzle ROI. Two extraction strategies will be compared: hand-crafted descriptors such as grey-level co-occurrence matrices (GLCM) that encode spatial frequency and directionality, and learned representations from a convolutional encoder trained on the same ROI crops. Paper 2 will report which strategy yields higher classification accuracy and whether the two capture complementary information. Fusion can be implemented at different levels of the processing pipeline, and the choice affects both accuracy and interpretability. Early fusion concatenates raw or lightly processed signals from multiple modalities before feature extraction, preserving cross-modal correlations but increasing input dimensionality (Petrich et al., 2021). Mid-level fusion extracts features from each modality through separate encoder branches and merges them at an intermediate layer, allowing each branch to learn modality-specific representations before combination. Hybrid Convolutional Neural Networks (CNNs) following this approach have achieved high accuracy in laser-directed energy deposition without manual feature engineering (Chen et al., 2024). Late fusion trains independent classifiers per modality and combines their outputs at the decision level, which is simpler to implement but discards inter-modal feature interactions. Paper 2 will evaluate early and mid-level fusion to determine whether cross-modal feature interaction improves state estimation. For each configuration, classification accuracy is reported per modality, per labeling strategy, and per process state. The time at which each modality first flags a state transition is also recorded for every induced-failure sequence to compare detection timing between texture and geometry. The raw images recorded during Paper 1’s volumetric prints are available as supplementary test data.

Metrics:

- *Classification F1 per process state:* standard metric; reported separately for each modality, fusion strategy, and labeling strategy.
- *Detection time gap (s):* time difference between texture-based and geometry-based first detection of the same state transition; reported per failure mode. Defined here because existing polymer literature hypothesizes that texture precedes geometry but no study has measured the gap.
- *Pipeline latency (ms/frame):* standard computational metric; confirms whether each modality and fusion configuration meets the real-time constraint from Paper 1.

Paper 3: Cross-Polymer Transfer

Paper 2 establishes which modality and fusion strategy work best, while this paper treats that pipeline as a fixed object and studies how it behaves when the material changes. The question is not which features are most accurate, but whether the learned representations remain valid across polymers with different rheological behavior, and what drives failure when they do not.

Material selection: The source material is LX175 PLA, used throughout Papers 1 and 2. At least two target materials with distinct flow behavior are selected: one semi-crystalline polymer with modified rheology, such as PLA compounded with short carbon fiber, which raises melt viscosity and alters die swell, and one amorphous polymer, such as ABS, which softens gradually above T_g rather than exhibiting

a sharp melting transition. This selection is driven by the rheological contrast discussed in the Background: semi-crystalline and amorphous polymers respond differently to the same thermal input.

Zero-shot evaluation: The best-performing perception pipeline from Paper 2 (modality, fusion strategy, and classifier) is applied to each target material without any retraining. The induced-failure experiment protocol from Paper 2 is repeated on each new material to generate comparable test data. Classification performance is measured per process state and per modality to diagnose where transfer breaks down. Not all process states observed in PLA may occur in the target materials, and new failure modes may appear. Both outcomes are reported.

Few-shot adaptation: Starting from the zero-shot baseline, incrementally larger subsets of labeled data from the target material are used to fine-tune the classifier. An adaptation curve is constructed for each target material, plotting F1 recovery against the number of labeled samples. This answers a practical question: when switching to a new polymer, how many calibration prints are needed before the perception module becomes reliable again.

Transfer failure analysis: All images from the three materials are passed through the same encoder, producing embeddings in a shared feature space. These embeddings are projected to two dimensions via dimensionality reduction (e.g., t-SNE) to visualize whether the same process states cluster together across materials or separate by material. If PLA sharkskin and ABS sharkskin land in different regions of the embedding space, the representation is not material-invariant for that state. This analysis is repeated per modality to determine whether geometry features, texture features, or fused features produce the most material-invariant representations. The goal is to identify whether transfer failure is driven primarily by the choice of visual features, by physical differences between the materials, or by the classifier architecture.

Metrics:

- *Zero-shot F1 drop:* difference in F1 between source and target material, reported per process state and per modality. Standard transfer learning evaluation.
- *Samples to recover performance:* number of labeled target-material samples required to restore F1 to a specified percentage of the source-material baseline. Defined here to quantify adaptation cost, which has practical relevance for deployment but is not a standardized metric in AM literature.
- *Embedding alignment:* quantitative assessment, like silhouette score of whether same-state clusters from different materials overlap in the learned feature space. Standard practice in domain adaptation research for evaluating representation transferability.

The volumetric test geometries are printed in each target material under closed-loop control using the adapted perception pipeline. Together with the PLA prints from Paper 1, these constitute the final set of architectural demonstrators: the same design intent realized across three materials, showing whether geometric fidelity is maintained when the framework adapts to different polymers.

Timeline

- **Phase 1: Closed-Loop Fidelity (Spring – Summer 2026)**
 - Milestone: Finish Paper 1.
 - Target: Submit to Robotics and Computer-Integrated Manufacturing (RCIM) or IEEE CASE.
- **Phase 2: Texture vs. Geometry (Fall 2026 – Spring 2027)**
 - Milestone: Finish Paper 2. Submit Paper 1.
 - Target: Submit to Additive Manufacturing or Journal of Manufacturing Systems.
- **Phase 3: Cross-Polymer Transfer (Summer – Fall 2027)**
 - Milestone: Finish Paper 3. Submit Paper 2.
 - Target: Submit to Automation in Construction or Virtual and Physical Prototyping.

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